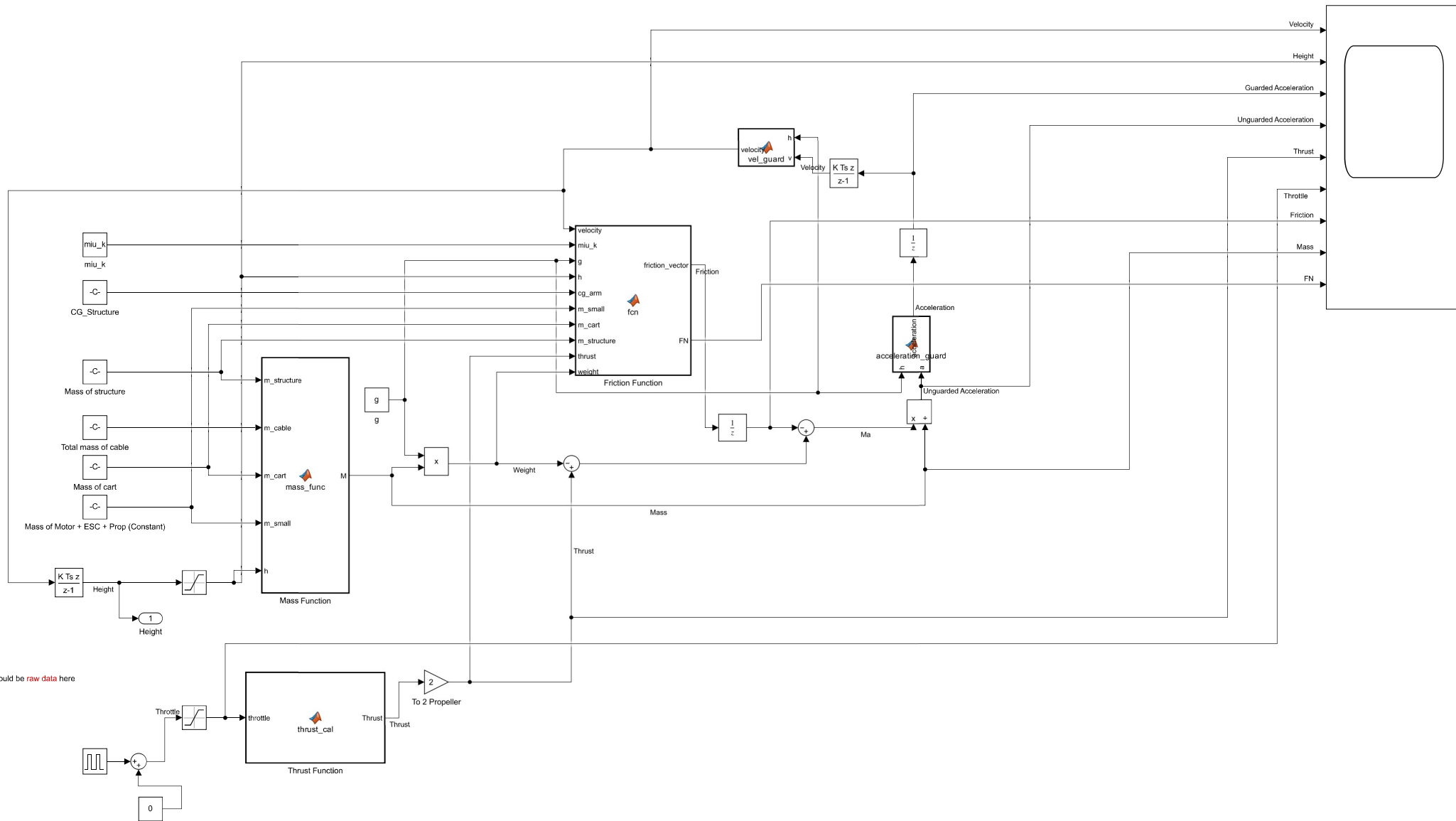




Throttle could be raw data here

```

function [friction_vector, FN]= fcn(velocity, miu_k, g, h, cg_arm, m_small, m_cart, m_structure, thrust, weight)
% 1. Initialize output to satisfy MATLAB Coder requirements
friction_vector = 0;

m_cable = 0.095 * h + 0.053656;
y_pad = 0.171;
F_max = 2.5506;
x_thrust = cg_arm;
cg_cart = (7.889 + 27) * 0.001;

% Resultant driving force
resultant_force = thrust - weight;

% Normal force between upper and lower pads
FN = 2 * (thrust * x_thrust - (m_structure + 2 * m_small) * g * cg_arm ...
    - (m_cart + m_cable) * g * cg_cart) / y_pad;
FN = max(FN, 0);

F_dynamic = FN * miu_k;
v_eps = 0.001;

% 2. Calculate a damping gain for the stiction zone
% This will actively pull the residual velocity exactly to 0
kv = F_max / v_eps;

if (abs(velocity) <= v_eps) && (abs(resultant_force) < F_max)
    % Apply balancing force PLUS the active damping
    f_static = -resultant_force - (kv * velocity);

    % Clamp it so the virtual damping doesn't accidentally exceed the static limit
    friction_vector = max(min(f_static, F_max), -F_max);

elseif (abs(velocity) <= v_eps) && (abs(resultant_force) >= F_max)
    % Static friction limit exceeded (Breakaway)
    friction_vector = -F_max * sign(resultant_force);

else
    % Dynamic friction, opposite to velocity
    friction_vector = -F_dynamic * sign(velocity);
end
end

```

```
function acceleration = acceleration_guard(h,a)

if h==3
    acceleration = min(a,0);
elseif h==0
    acceleration = max(a,0);
else
    acceleration = a;
end
```

```
function velocity = vel_guard(h,v)
```

```
if h>=3
    velocity = 0;
elseif h<=0
    velocity = 0;
else
    velocity = v;
end
```

```
end
```

```
function M = mass_func(m_structure,m_cable,m_cart,m_small,h)
M = (2 * (m_small)) + m_cart + m_structure + m_cable*(0.5648 + h);
```

```
function Thrust = thrust_cal(throttle)

if throttle >= 0 && throttle <= 79.0425

    Thrust = ...
        -1.02178722e-11 * throttle^7 + ...
        3.31547781e-09 * throttle^6 + ...
        -4.23315827e-07 * throttle^5 + ...
        2.75535168e-05 * throttle^4 + ...
        -9.84949443e-04 * throttle^3 + ...
        1.84147041e-02 * throttle^2 + ...
        -2.92219275e-02 * throttle + ...
        5.09621010e-02;

elseif throttle > 79.0425 && throttle <= 100

    Thrust = 7.4300463179;

else

    error('Invalid throttle input');

end

end
```